Conformal Field Theory for Babies

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1 Introduction

These lectures are based on the CFT seminars in HKUST(gz) from March to July.

1.1 Conforaml symmetries in d-dimension

Nowadays, we know that our world are composed by elementary particles, like quarks, photons, electrons, etc, quarks and gluons composed protons and netrons, together with electrons are the building block of nucleus. What is amazing in Quantum field theory is that these elementary particles are described by the irreducible representations of the symmetry group of Minikovski space-time, that is how quantum theory is combined with special relativity. Quantum field theory is a huge success such that almost all phenomons observed by human in nature can be described by this theory. However, people find that many problems in QFT are very hard to solve, and only a few models are exactly solvable, they may caused by more symmetries in this system. One pheonmon in statistical physics is the critical phenomena, which tells us that at critical point, our systerm will have infinte correlation lenth, which means the sysmtem is scale invariant.

In 1970s, what surprised thing Alexander Polyakov find is that critical systems will have more symmetry than scale invariant(also Poincaré group), called conformal symmetry. The most well known physical model – Ising model whose 2D version has a phase transition from ordered phase(ferromagnetic) to disordered phase(paramagnetic), and at the critical point(also called the self-dual point if you are familiar with Ising model), the system has conformal symmetry, and the theory is well understood, now is called Ising CFT.

First let me introduce the formal definition of conformal transformation and explore the structure of the conformal group. We consider the space $\boldsymbol{R}^d$ with flat Riemannian metric $g_{\mu\nu}$, $g_{\mu\nu} = diag(1, -1, ..., -1)$ if it is Minikovski space and $g_{\mu\nu} = diag(1, 1, ..., 1)$ if it is Euclidean space, the Riemannian metric tells us how to measure the infinitesimal distance: $ds^2 = g_{\mu\nu}dx^\mu dx^\nu$. A coordinate transformation $x \to x' = x'(x)$ changes the original metric to a new metric with respect to the new coordinate system:

$$g_{\mu\nu}(x) \to g'_{\mu\nu}(x') = \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}g_{\alpha\beta}(x), \tag{1}$$

this formula can be derived through the chain rule: the distance at point x is

$$ds^2 = g_{\alpha\beta}(x)dx^\alpha dx^\beta = g_{\alpha\beta}(x)\frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}dx'^\mu dx'^\nu. \tag{2}$$

A conformal transformation is a coordinate transformation that preserves the Riemannian metric up to a scale factor, that is

$$g'_{\mu\nu}(x') \propto \Omega(x)g_{\mu\nu}(x), \tag{3}$$

where $\Omega(x)$ is a function of the coordinate x, we can see that the Poincaré transformations are the subset of the conformal transformations with the metric invariant. Note that conformal transformation preserves angles, recall the definition of the angle between two vectors $\frac{v\cdot w}{\sqrt{v\cdot v}\sqrt{w\cdot w}}$.

All conformal transformations composed a group called Conformal group, which is a Lie group. In physics, it is instructive to study the infinitesimal generator of the conformal group, now considering a infinitesimal coordinate transformation $x^\mu \to x^\mu + \varepsilon^\mu(x)$, under this transformation, we can derive the new metric tensor as follows:

$$
\begin{aligned}
g'_{\mu\nu}(x') &= \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu} g_{\alpha\beta}(x) = \frac{\partial(x'^\alpha - \varepsilon^\alpha)}{\partial x'^\mu}\frac{\partial\big(x'^\beta - \varepsilon^\beta\big)}{\partial x'^\nu} g_{\alpha\beta}(x) \\
&= \left(\delta^\alpha_\mu - \frac{\partial \varepsilon^\alpha}{\partial x'^\mu}\right)\cdot\left(\delta^\beta_\nu - \frac{\partial \varepsilon^\beta}{\partial x'^\nu}\right) g_{\alpha\beta}(x) \\
&= \delta^\alpha_\mu \delta^\beta_\nu g_{\alpha\beta}(x) - \frac{\partial \delta^\alpha_\mu g_{\alpha\beta}\varepsilon^\beta}{\partial x'^\nu} - \frac{\partial \delta^\beta_\nu g_{\alpha\beta}\varepsilon^\alpha}{\partial x'^\mu} + o\big(|\varepsilon|^2\big) \\
&= g_{\mu\nu}(x) - \big(\partial_\nu \varepsilon_\mu + \partial_\mu \varepsilon_\nu\big).
\end{aligned} \tag{4}
$$

To satisfy the condition Equation 3, we need $\partial_\nu \varepsilon_\mu + \partial_\mu \varepsilon_\nu = \lambda g_{\mu\nu}(x)$, using $g^{\mu\nu}(x)$ to contract both sides, we get

$$
g^{\mu\nu}\big(\partial_\nu \varepsilon_\mu + \partial_\mu \varepsilon_\nu\big) = \partial_\nu \varepsilon^\nu + \partial_\mu \varepsilon^\mu = 2\partial\cdot\varepsilon, \lambda g^{\mu\nu} g_{\mu\nu} = d\cdot\lambda \Rightarrow \lambda = \frac{2}{d}\partial\cdot\varepsilon. \tag{5}
$$

So that we have $\Omega(x) = 1 + \frac{2}{d}\partial\cdot\varepsilon$, and

$$
\partial_\mu \varepsilon_\nu + \partial_\nu \varepsilon_\mu = \frac{2}{d}(\partial\cdot\varepsilon) g_{\mu\nu}. \tag{6}
$$

Next we will study the explicit form of conformal transformations in d dimension, taking the derivative on both sides of Equation 6, we get:

$$
\begin{aligned}
&\partial_\rho\big(\partial_\mu \varepsilon_\nu + \partial_\nu \varepsilon_\mu\big) = \frac{2}{d}\partial_\rho\big(\partial\cdot\varepsilon g_{\mu\nu}\big), \\
&\partial_\rho\partial_\mu \varepsilon_\nu + \partial_\rho\partial_\nu \varepsilon_\mu = \frac{2}{d} g_{\mu\nu}\partial_\rho(\partial\cdot\varepsilon) + \frac{2}{d}(\partial\cdot\varepsilon)\partial_\rho g_{\mu\nu} = \frac{2}{d} g_{\mu\nu}\partial_\rho(\partial\cdot\varepsilon),
\end{aligned} \tag{7}
$$

permute the indices we get:

$$
\partial_\nu\partial_\rho \varepsilon_\mu + \partial_\mu\partial_\rho \varepsilon_\nu = \frac{2}{d} g_{\rho\mu}\partial_\nu(\partial\cdot\varepsilon), \tag{8}
$$

$$
\partial_\mu\partial_\nu \varepsilon_\rho + \partial_\nu\partial_\mu \varepsilon_\rho = \frac{2}{d} g_{\nu\rho}\partial_\mu(\partial\cdot\varepsilon). \tag{9}
$$

Note that the second derivative operator is symmetric, Equation 9 becomes

$$
2\partial_\mu\partial_\nu \varepsilon_\rho = \frac{2}{d} g_{\nu\rho}\partial_\mu(\partial\cdot\varepsilon), \tag{10}
$$

and also the second formula of Equation 7 becomes:

$$
\partial_\nu\partial_\rho \varepsilon_\mu + \partial_\mu\partial_\rho \varepsilon_\nu = \frac{2}{d} g_{\mu\nu}\partial_\rho(\partial\cdot\varepsilon). \tag{11}
$$

$(10) + (8) - (11)$ gives

$$
2\partial_\mu\partial_\nu \varepsilon_\rho = \frac{2}{d} g_{\rho\mu}\partial_\nu(\partial\cdot\varepsilon) + \frac{2}{d} g_{\nu\rho}\partial_\mu(\partial\cdot\varepsilon) - \frac{2}{d} g_{\mu\nu}\partial_\rho(\partial\cdot\varepsilon), \tag{12}
$$

using $g^{\mu\nu}$ to contract Equation 12, we get

$$
\begin{aligned}
2\partial^2\varepsilon_\rho &= \frac{2}{d}g_\rho^\nu\partial_\nu(\partial\cdot\varepsilon) + \frac{2}{d}g_\rho^\mu\partial_\mu(\partial\cdot\varepsilon) - 2\partial\rho(\partial\cdot\varepsilon) \\
\Rightarrow \partial^2\varepsilon_\rho &= \left(\frac{2}{d}-1\right)\partial_\rho(\partial\cdot\varepsilon),
\end{aligned}
\tag{13}
$$

also you can write it as

$$
\partial^2\varepsilon_\mu = \left(\frac{2}{d}-1\right)\partial_\mu(\partial\cdot\varepsilon), \tag{14}
$$

take ∂_ν on both sides of Equation 14, and take ∂^2 on both sides of Equation 6, we get:

$$
\begin{aligned}
\partial^2\partial_\nu\varepsilon_\mu &= \left(\frac{2}{d}-1\right)\partial_\nu\partial_\mu(\partial\cdot\varepsilon), \\
\partial^2\partial_\mu\varepsilon_\nu + \partial^2\partial_\nu\varepsilon_\mu &= \frac{2}{d}g_{\mu\nu}\partial^2(\partial\cdot\varepsilon),
\end{aligned}
\tag{15}
$$

Equation 15 gives us:

$$
g_{\mu\nu}\partial^2(\partial\cdot\varepsilon) = (2-d)\partial_\mu\partial_\nu(\partial\cdot\varepsilon), \tag{16}
$$

use $g^{\mu\nu}$ to contract both sides, we have other formula:

$$
(d-1)\partial^2(\partial\cdot\varepsilon) = 0. \tag{17}
$$

For $d=1$, the above equations impose no constraint on the transformation, so that any smmoth transformations are conformal in one dimension. And $d=2$ is a very special case which we will mainly focus on it, and we will come back to this case later. For $d>2$ cases, from Equation 17 and Equation 16 we can see that $\partial_\mu\partial_\nu(\partial\cdot\varepsilon)=0$, which means $\partial\cdot\varepsilon$ is at most linear in the coordinates, so that we can write:

$$
\partial\cdot\varepsilon = A + B_\mu x^\mu, \tag{18}
$$

subsititue Equation 18 into Equation 12 we get

$$
\partial_\mu\partial_\nu\varepsilon_\rho = \frac{1}{d}(g_{\rho\mu}B_\nu + g_{\nu\rho}B_\mu - g_{\mu\nu}B_\rho), \tag{19}
$$

which means every second order derivative $\partial_\mu\partial_\nu\varepsilon_\rho$ is a constant, so ε_μ is at most quadratic in the coordinates, and the general form of ε_μ is

$$
\varepsilon_\mu = a_\mu + c_{\mu\nu}x^\nu + f_{\mu\nu\rho}x^\nu x^\rho \text{ where } \quad f_{\mu\nu\rho} = f_{\mu\rho\nu}. \tag{20}
$$

First, we look at the zeroth order in x, i.e., a_μ terms, obviously there are no constraints in these terms, i.e., a_μ are free variables, these correspond to ordinary translations independent of x. Now focusing on the linear terms, substituing Equation 20 into Equation 6, after doing derivative, linear terms become constant terms, and these constant terms should satisfy:

$$
c_{\mu\nu} + c_{\nu\mu} = \frac{2}{d}c^\rho{}_\rho g_{\mu\nu}, \tag{21}
$$

viewing $c_{\mu\nu}$ as a matrix, this shows when $\mu\neq\nu$, we have $c_{\mu\nu}=-c_{\nu\mu}$, when $\mu=\nu$, we have $c_{\mu\mu}=\alpha g_{\mu\mu}$, where α is a constant so that we can decompose $c_{\mu\nu}$ into the sum of pure trace part and anti-symmetric part:

$$c_{\mu\nu} = \alpha g_{\mu\nu} + m_{\mu\nu}, \tag{22}$$

and $m_{\mu\nu} = -m_{\nu\mu}$, and the pure trace part correspond to dilation, the anti-symmetric part correspond to rotation. Finally, substituting the quadratic terms $\varepsilon_\mu = f_{\mu\nu\rho}x^\nu x^\rho$ into Equation 12, we get the left hand side is:

$$\begin{aligned}\partial_\mu\partial_\nu\varepsilon_\rho &= \partial_\mu\partial_\nu f_{\rho\mu'\nu'}x^{\mu'}x^{\nu'} = \partial_\mu\left(f_{\rho\mu'\nu'}\delta^{\mu'}_\nu x^{\nu'} + f_{\rho\mu'\nu'}\delta^{\nu'}_\nu x^{\mu'}\right)\\ &= f_{\rho\mu'\nu'}\delta^{\mu'}_\nu\delta^{\nu'}_\mu + f_{\rho\mu'\nu'}\delta^{\nu'}_\nu\delta^{\mu'}_\mu = 2f_{\rho\mu\nu},\end{aligned} \tag{23}$$

and the right hand side is:

$$\begin{aligned}\frac{1}{d}g_{\rho\mu}\partial_\nu(\partial_\sigma\varepsilon^\sigma) &= \frac{1}{d}g_{\rho\mu}\partial_\nu\left(\partial_\sigma f^\sigma{}_{\mu'\nu'}x^{\mu'}x^{\nu'}\right) = \frac{1}{d}g_{\rho\mu}\partial_\nu\left(f^\sigma{}_{\mu'\nu'}\delta^{\mu'}_\sigma x^{\nu'} + f^\sigma{}_{\mu'\nu'}\delta^{\nu'}_\sigma x^{\mu'}\right)\\ &= \frac{1}{d}g_{\rho\mu}\left(f^\sigma{}_{\mu'\nu'}\delta^{\mu'}_\sigma\delta^{\nu'}_\nu + f^\sigma{}_{\mu'\nu'}\delta^{\nu'}_\sigma\delta^{\mu'}_\nu\right) = \frac{2}{d}g_{\rho\mu}f^\sigma{}_{\sigma\nu},\end{aligned} \tag{24}$$

so that we have

$$f_{\rho\mu\nu} = g_{\rho\mu}b_\nu + g_{\nu\rho}b_\mu - g_{\mu\nu}b_\rho \text{ where } \quad b_\mu \equiv \frac{1}{d}c^\sigma{}_{\sigma\mu}, \tag{25}$$

or you can permute the indices and get:

$$f_{\mu\nu\rho} = g_{\rho\mu}b_\nu + g_{\mu\nu}b_\rho - g_{\nu\rho}b_\mu. \tag{26}$$

In this case, we can simplify the form of ε_μ:

$$\begin{aligned}\varepsilon_\mu &= f_{\mu\nu\rho}x^\nu x^\rho = g_{\rho\mu}b_\nu x^\nu x^\rho + g_{\mu\nu}b_\rho x^\nu x^\rho - g_{\nu\rho}b_\mu x^\nu x^\rho\\ &= x_\mu b_\nu x^\nu + x_\mu b_\rho x^\rho - b_\mu x_\nu x_\rho,\end{aligned} \tag{27}$$

so that the corresponding infinitesimal transoformation is:

$$x'^\mu = x^\mu + 2(\boldsymbol{x}\cdot\boldsymbol{b})x^\mu - b^\mu\boldsymbol{x}^2, \tag{28}$$

which is call the *special conformal transformation*(SCT), the finite form for this transformation is:

$$x^\mu \longmapsto \frac{x^\mu - b^\mu\boldsymbol{x}^2}{1 - 2\boldsymbol{b}\cdot\boldsymbol{x} + \boldsymbol{b}^2\boldsymbol{x}^2}, \tag{29}$$

an elegant way to see how the SCTs are derived is to decompse it into inversions and translations, actually, SCT = Inversion ∘ Translation ∘ Inversion:

$$\begin{aligned}&\text{Inversion:} && x^\mu \longmapsto \frac{x^\mu}{\boldsymbol{x}^2}\\ &\text{Translation:} && \frac{x^\mu}{\boldsymbol{x}^2} \longmapsto \frac{x^\mu}{\boldsymbol{x}^2} - b^\mu\\ &\text{Inversion:} && \frac{x^\mu}{\boldsymbol{x}^2} - b^\mu \longmapsto \frac{\frac{x^\mu}{\boldsymbol{x}^2} - b^\mu}{\left(\frac{x^\mu}{\boldsymbol{x}^2} - b^\mu\right)\left(\frac{x_\mu}{\boldsymbol{x}^2} - b_\mu\right)} = \frac{x^\mu - b^\mu\boldsymbol{x}^2}{1 - 2\boldsymbol{b}\cdot\boldsymbol{x} + \boldsymbol{b}^2\boldsymbol{x}^2}\end{aligned}$$

1.2 Conformal symmetries in 2-dimension

As we can see in the constraint equations in section 1.1, $d = 2$ is very special case, which is our main focus in this seminar, and we also assume that $g_{\mu\nu} = \delta_{\mu\nu}$, i.e., we are treating Euclidean space-time. Now the Equation 6 becomes the famous Cauchy-Riemann equation:

$$\partial_1\varepsilon_1 = \partial_2\varepsilon_2, \quad \partial_1\varepsilon_2 = -\partial_2\varepsilon_1. \tag{30}$$

So we introduce complex coordinates:

$$z = x^1 + ix^2, \quad \overline{z} = x^1 - ix^2. \tag{31}$$

Here we should mention that the global conformal transformations of the complex plan are entire functions, and you can prove that only linear functions satisfy the condition, when we expand the complex plane into $\mathbb{C} \cup \{\infty\}$, then the global conformal transformations are the so called Mobius transoformations which have the form:

$$f(z) = \frac{az+b}{cz+d} \quad \text{with} \quad ad - bc = 1, \tag{32}$$

where a, b, c and d are complex numbers, written a, b, c, d as matrix we have

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2;\mathbb{C}). \tag{33}$$

It seems that the global conformal transoformations in 2D are very constrained, however, the local conformal symmetries have richer structure comparing to high dimensions, and from the physical viewpoint, local properties are more useful than the global properties. Now we will further study the structure of the local algebra of infinitesimal conformal transformations, they can be expressed as

$$z' = z + \varepsilon(z) \quad \text{where } \varepsilon(z) = \sum_{-\infty}^{\infty} c_n z^{n+1}, \tag{34}$$

$\varepsilon(z)$ can be expressed as Laurent series confirm this infinitesimal transformation is conformal, now the effect of the transformation on a field $\phi(z,\overline{z})$ is

$$\phi'(z',\overline{z}') = \phi(z,\overline{z}) = \phi(z',\overline{z}') - \varepsilon(z')\partial_{z'}\phi(z',\overline{z}') - \overline{\varepsilon}(\overline{z}')\partial_{\overline{z}'}\phi(z',\overline{z}'), \tag{35}$$

we introduce the basis

$$l_n = -z^{n+1}\partial_z, \quad \bar{l}_n = -\overline{z}^{n+1}\partial_{\overline{z}} \tag{36}$$

so that we have

$$\delta\phi = \sum_n \{c_n l_n \phi(z,\overline{z}) + \overline{c}_n \bar{l}_n \phi(z,\overline{z})\}. \tag{37}$$

These generators obey the following commutation relations:

$$\begin{aligned} [l_n, l_m] &= z^{n+1}\partial_z(z^{m+1}\partial_z) - z^{m+1}\partial_z(z^{n+1}\partial_z) = (n-m)l_{n+m}, \\ \left[\bar{l}_n, \bar{l}_m\right] &= \overline{z}^{n+1}\partial_{\overline{z}}(\overline{z}^{m+1}\partial_{\overline{z}}) - \overline{z}^{m+1}\partial_{\overline{z}}(\overline{z}^{n+1}\partial_{\overline{z}}) = (n-m)l_{n+m}, \\ \left[l_n, \bar{l}_m\right] &= z^{n+1}\partial_z(\overline{z}^{m+1}\partial_{\overline{z}}) - \overline{z}^{m+1}\partial_{\overline{z}}(z^{n+1}\partial_z) = 0. \end{aligned} \tag{38}$$

From the commutation rules we can see that the local conformal algebra is the direct sum of two isomorphic algebras $\mathcal{A} \oplus \overline{\mathcal{A}}$, and it is called local because they are not well-defined globally in the extended complex plane. Now from the local conformal algebra we can learn more about the global conformal transformations, note that the holomorphic conformal transformations are generated by vector fields

$$v(z) = -\sum_n a_n l_n = \sum_n a_n z^{n+1}\partial_z, \tag{39}$$

if it is well-defined globally, it means no singularity of $v(z)$ as $z \to 0$, which needs $a_n \neq 0$ only for $n \geq -1$, for demonstrating the behaviour of $v(z)$ as $z \to \infty$, we perform the transformation $z = -\frac{1}{\omega}$, then

$$v(z) = \sum_n a_n \left(-\frac{1}{\omega}\right)^{n+1} \left(\frac{dz}{d\omega}\right)^{-1} \partial_\omega = \sum_n a_n \left(-\frac{1}{\omega}\right)^{n-1} \partial_\omega, \tag{40}$$

no singularity as $\omega \to 0$ means $a_n \neq 0$ only for $n \leq 1$. So that globally defined holomorphic conformal transformations are generated by $\{l_0, l_{-1}, l_1\}$, and also the anti-holomorphic conformal transformations are generated by $\{\bar{l}_0, \bar{l}_{-1}, \bar{l}_1\}$. Moreover, $l_{-1} = -\partial_z$ and $\bar{l}_{-1} = -\partial_{\bar{z}}$ are the generators of the translation, $l_0 + \bar{l}_0 = -z\partial_z - \bar{z}\partial_{\bar{z}} \propto -x\partial_x - y\partial_y$ is the generator of dilation, $i\left(l_0 - \bar{l}_0\right) \propto y\partial_x - x\partial_y$ is the generator of rotation, and $l_1 = -z^2\partial_z, \bar{l}_1 = -\bar{z}^2\partial_{\bar{z}}$ are the generators of special conformal transformations.

1.3 Constraints of conformal invariance in d dimension

Now we come back to the case of an arbitrary number of dimensions $d = p + q$ and see how the conformal symmetries give us constraints on the N-point functions of a field theory. Recall the classical field theory, suppose we have an action

$$\mathcal{S} = \int d^d x \mathcal{L}(\Phi, \partial_\mu \Phi), \tag{41}$$

an active transformation affect both the postition and the fiedls:

$$\begin{aligned} \boldsymbol{x} &\longmapsto \boldsymbol{x}' \\ \Phi(\boldsymbol{x}) &\longmapsto \Phi'(\boldsymbol{x}') \end{aligned} \tag{42}$$

In theses transformations, the new position $\boldsymbol{x}'$ is a function of $\boldsymbol{x}$ and the new field $\Phi'(\boldsymbol{x}')$ is a function of $\Phi(\boldsymbol{x})$, i.e., $\Phi'(\boldsymbol{x}') = \mathcal{F}(\Phi(\boldsymbol{x}))$, under this transformation, the original action changes to

$$\begin{aligned} \mathcal{S}' &= \int d^d x' \mathcal{L}\big(\Phi'(\boldsymbol{x}'), \partial'_\mu \Phi(\boldsymbol{x}')\big) \\ &= \int d^d x' \mathcal{L}\big(\mathcal{F}(\Phi(\boldsymbol{x})), \partial'_\mu \mathcal{F}(\Phi(\boldsymbol{x}))\big) \\ &= \int d^d x \left|\frac{\partial \boldsymbol{x}'}{\partial \boldsymbol{x}}\right| \mathcal{L}\left(\mathcal{F}(\Phi(\boldsymbol{x})), \frac{\partial x^\nu}{\partial x'^\mu}\partial_\nu \mathcal{F}(\Phi(\boldsymbol{x}))\right), \end{aligned} \tag{43}$$

where $\left|\frac{\partial \boldsymbol{x}'}{\partial \boldsymbol{x}}\right|$ is the determinant of the Jacobian. In general, when we perfrom a transformation, like Lorentz transformation, we have the following form:

$$\begin{aligned} x'^\mu &= \Lambda^\mu_\nu x^\nu \\ \Phi(\Lambda \boldsymbol{x}) &= L_\Lambda \Phi(\boldsymbol{x}), \end{aligned} \tag{44}$$

L forms a representation of the transformation group. From the representation theory, it is easier to study the infinitesimal generator first, i.e. the Lie algebra representation, and exponetial it into the group representation. If we suppose for the moment that the fields are unaffected by the transformation, i.e., $\mathcal{F}(\Phi) = \Phi$, then the generators of the global conformal group are:

$$\boxed{\begin{aligned} \text{(translation)} \qquad & P_\mu &= -i\partial_\mu \\ \text{(dilation)} \qquad & D &= -ix^\mu\partial_\mu \\ \text{(rotation)} \qquad & L_{\mu\nu} &= i\big(x_\mu\partial_\nu - x_\nu\partial_\mu\big) \\ \text{(SCT)} \qquad & K_\mu &= -i\big(2x_\mu x^\nu\partial_\nu - \boldsymbol{x}^2\partial_\mu\big) \end{aligned}} \tag{45}$$

And these generators satisfies the following commutation relations:

$$\begin{aligned} \big[D, P_\mu\big] &= iP_\mu \\ \big[D, K_\mu\big] &= -iK_\mu \\ \big[K_\mu, P_\nu\big] &= 2i\big(g_{\mu\nu}D - L_{\mu\nu}\big) \\ \big[K_\rho, L_{\mu\nu}\big] &= i\big(g_{\rho\mu}K_\nu - g_{\rho\nu}K_\mu\big) \\ \big[P_\rho, L_{\mu\nu}\big] &= i\big(g_{\rho\mu}P_\nu - g_{\rho\nu}P_\mu\big) \\ \big[L_{\mu\nu}, L_{\rho\sigma}\big] &= i\big(g_{\nu\rho}L_{\mu\sigma} + g_{\mu\sigma}L_{\nu\rho} - g_{\mu\rho}L_{\nu\sigma} - g_{\nu\sigma}L_{\mu\rho}\big) \end{aligned} \tag{46}$$

From the QFT, we already know a field transforming under Poincaré group, i.e., those transformations generated by P_μ and $L_{\mu\nu}$, so that we need to characterize how D and K affect these fields. First, we consider the subgroup of Conformal group which leaves the point $x = 0$ invariant, i.e., those transformations generated by $L_{\mu\nu}, K_\mu$ and D , and suppose we have

$$L_{\mu\nu}\Phi(0) = S_{\mu\nu}\Phi(0),\, D\Phi(0) = \tilde{\Delta}\Phi(0),\, K_\mu\Phi(0) = \kappa_\mu\Phi(0), \tag{47}$$

then $S_{\mu\nu}, \tilde{\Delta}, \kappa_\mu$ forms a representation of the Conformal subalgebra, they also satisfies the commutation relation in Equation 46, that is

$$\begin{aligned} \big[\tilde{\Delta}, S_{\mu\nu}\big] &= 0, \\ \big[\tilde{\Delta}, \kappa_\mu\big] &= -i\kappa_\mu, \\ \big[\kappa_\nu, \kappa_\mu\big] &= 0, \\ \big[\kappa_\rho, S_{\mu\nu}\big] &= i\big(g_{\rho\mu}\kappa_\nu - g_{\rho\nu}\kappa_\mu\big), \\ \big[S_{\mu\nu}, S_{\rho\sigma}\big] &= i\big(g_{\nu\rho}S_{\mu\sigma} + g_{\mu\sigma}S_{\nu\rho} - g_{\mu\rho}S_{\nu\sigma} - g_{\nu\sigma}S_{\mu\rho}\big). \end{aligned} \tag{48}$$

If we demand $S_{\mu\nu}$ are irreducible representations of the Lorentz group, then by Schur's lemma, any matrix that commutes with all $S_{\mu\nu}$ must be proportion to the identity, so that we have

$$\tilde{\Delta} \propto I \quad \Rightarrow -iK_\mu = 0. \tag{49}$$

From this we can conclude that we can write $\tilde{\Delta} = -i\Delta I$ and SCTs act trivially on field labels, non-trivially on spatial labels. After deriving the action at point $x = 0$, how can we derive the general formula for arbitrary point x? This can be done by the so called Hausdorff formula:

$$e^{-A}Be^{A} = B + [B, A] + \frac{1}{2!}[[B, A], A] + \frac{1}{3!}[[[B, A], A], A] + \cdots, \tag{50}$$

and

$$
\begin{aligned}
e^{ix^\mu P_\mu} D e^{-ix^\mu P_\mu} &= D + \left[D, -ix^\mu P_\mu\right] + \cdots = D + x^\mu P_\mu, \\
e^{ix^\nu P_\nu} K_\mu e^{-ix^\nu P_\nu} &= K_\mu + \left[K_\mu, -ix^\nu P_\nu\right] + \frac{1}{2!}\left[\left[K_\mu, -ix^\nu P_\nu\right], -ix^\nu P_\nu\right] + \cdots \\
&= K_\mu + 2\left(g_{\mu\nu} x^\nu D - L_{\mu\nu} x^\nu\right) \\
&= K_\mu + 2x_\mu D - 2x^\nu L_{\mu\nu} + 2x_\mu (x^\nu P_\nu) - \boldsymbol{x}^2 P_\mu,
\end{aligned}
\tag{51}
$$

so that the fields are transformed as:

$$
\begin{aligned}
D\Phi(\boldsymbol{x}) &= \left(-ix^\mu \partial_\mu + \tilde{\Delta}\right)\Phi(x) \\
K_\mu \Phi(\boldsymbol{x}) &= \left(\kappa_\mu + 2x_\mu \tilde{\Delta} - 2x^\nu S_{\mu\nu} - 2ix_\mu x^\nu \partial_\nu + i\boldsymbol{x}^2 \partial_\mu\right)\Phi(\boldsymbol{x}).
\end{aligned}
\tag{52}
$$

Now suppose the field is spinless field, then a finite dilation act as follows:

$$
e^{i\alpha D}\phi(\boldsymbol{x}) = e^{\alpha\Delta}\phi(e^\alpha \boldsymbol{x}),
\tag{53}
$$

if we identify $e^\alpha = \lambda$, then we have:

$$
\begin{aligned}
\boldsymbol{x} &\longmapsto \lambda \boldsymbol{x} \\
\phi(\boldsymbol{x}) &\longmapsto \lambda^\Delta \phi(\boldsymbol{x})
\end{aligned}
\tag{54}
$$

We can conclude that these fields under global conformal transformations, $x \to x'$ transform according to

$$
\phi_j(\boldsymbol{x}) \to \left|\frac{\partial \boldsymbol{x}'}{\partial \boldsymbol{x}}\right|^{\frac{\Delta_j}{d}} \phi_j(\boldsymbol{x}').
\tag{55}
$$

A field transforming liked the above is called quasi-primary field, and for quasi-primary field the correlation functions satisfy:

$$
\langle \phi_1(\boldsymbol{x}_1) \cdots \phi_n(\boldsymbol{x}_n) \rangle = \left|\frac{\partial \boldsymbol{x}'}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x}=\boldsymbol{x}_1}^{\frac{\Delta_1}{d}} \cdots \left|\frac{\partial \boldsymbol{x}'}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x}=\boldsymbol{x}_n}^{\frac{\Delta_n}{d}} \langle \phi_1(\boldsymbol{x}_1') \cdots \phi_n(\boldsymbol{x}_n') \rangle
\tag{56}
$$

this property imposes severe restrications on 2- and 3-point functions of quasi-primary field. First, if we specialize to scale transformation $\boldsymbol{x} \to \lambda \boldsymbol{x}$ we obtain

$$
\langle \phi_1(\boldsymbol{x}_1)\phi_2(\boldsymbol{x}_2) \rangle = \lambda^{\Delta_1 + \Delta_2} \langle \phi_1(\lambda \boldsymbol{x}_1)\phi_2(\lambda \boldsymbol{x}_2) \rangle,
\tag{57}
$$

because in this case $\left|\frac{\partial \boldsymbol{x}'}{\partial \boldsymbol{x}}\right| = \lambda^d$, and the correaltion function are invariant under rotation and translation, translation invariance tells us that correaltion function should only depend on the differences $x_i - x_j$, and rotation invariance restricts it to $|\boldsymbol{x}_1 - \boldsymbol{x}_2|$, so that we must have

$$
\langle \phi_1(\boldsymbol{x}_1)\phi_2(\boldsymbol{x}_2) \rangle = f(|\boldsymbol{x}_1 - \boldsymbol{x}_2|),
\tag{58}
$$

combined with Equation 57, invariance under dilation gives us:

$$
\langle \phi_1(\boldsymbol{x}_1)\phi_2(\boldsymbol{x}_2) \rangle = \frac{C_{12}}{|\boldsymbol{x}_1 - \boldsymbol{x}_2|^{\Delta_1 + \Delta_2}}.
\tag{59}
$$

The last restriction is from the special conformal transformations, remind that special conformal transformation has the form:

$$
\begin{aligned}
x^\mu \longmapsto x'^\mu &= \frac{x^\mu - b^\mu \boldsymbol{x}^2}{1 - 2\boldsymbol{b}\cdot\boldsymbol{x} + \boldsymbol{b}^2\boldsymbol{x}^2}, \\
\left|\frac{\partial \boldsymbol{x}'}{\partial \boldsymbol{x}}\right| &= \frac{1}{(1 - 2\boldsymbol{b}\cdot\boldsymbol{x} + \boldsymbol{b}^2\boldsymbol{x}^2)^d},
\end{aligned} \tag{60}
$$

so the Equation 56 implies:

$$
\frac{C_{12}}{|\boldsymbol{x}_1 - \boldsymbol{x}_2|^{\Delta_1+\Delta_2}} = \frac{C_{12}}{\gamma_1^{\Delta_1}\gamma_2^{\Delta_2}} \frac{(\gamma_1\gamma_2)^{\frac{\Delta_1+\Delta_2}{2}}}{|\boldsymbol{x}_1 - \boldsymbol{x}_2|^{\Delta_1+\Delta_2}} \quad \text{where } \gamma_i = 1 - 2\boldsymbol{b}\cdot\boldsymbol{x}_i + \boldsymbol{b}^2\boldsymbol{x}_i^2. \tag{61}
$$

The main difficulty for deriving Equation 61 is to show $(\boldsymbol{x}_1' - \boldsymbol{x}_2')^2 = \frac{(\boldsymbol{x}_1 - \boldsymbol{x}_2)^2}{\gamma_1\gamma_2}$, so we show it explicitly

$$
\boldsymbol{x}_1' - \boldsymbol{x}_2' = \frac{(\gamma_2 x_1^\mu - \gamma_2 b^\mu \boldsymbol{x}_1^2 - \gamma_1 x_2^\mu + \gamma_1 b^\mu \boldsymbol{x}_2^2)}{\gamma_1\gamma_2}, \tag{62}
$$

let $\beta^\mu = \gamma_2 x_1^\mu - \gamma_2 b^\mu \boldsymbol{x}_1^2 - \gamma_1 x_2^\mu + \gamma_1 b^\mu \boldsymbol{x}_2^2$, then we want to compute

$$
(\boldsymbol{x}_1' - \boldsymbol{x}_2')^2 = \frac{\beta^\mu \beta_\mu}{\gamma_1^2\gamma_2^2}, \tag{63}
$$

$\beta^\mu\beta_\mu$ has 16 terms as follows:

$$
\begin{aligned}
&\gamma_2^2\boldsymbol{x}_1^2 - \gamma_2^2(\boldsymbol{b}\cdot\boldsymbol{x}_1)\boldsymbol{x}_1^2 - \gamma_1\gamma_2\boldsymbol{x}_1\cdot\boldsymbol{x}_2 + \gamma_1\gamma_2(\boldsymbol{b}\cdot\boldsymbol{x}_1)\boldsymbol{x}_2^2, \\
&-\gamma_2^2(\boldsymbol{b}\cdot\boldsymbol{x}_1)\boldsymbol{x}_1^2 + \gamma_2^2\boldsymbol{b}^2\boldsymbol{x}_1^4 + \gamma_1\gamma_2(\boldsymbol{b}\cdot\boldsymbol{x}_2)\boldsymbol{x}_1^2 - \gamma_1\gamma_2\boldsymbol{b}^2\boldsymbol{x}_1^2\boldsymbol{x}_2^2, \\
&-\gamma_1\gamma_2\boldsymbol{x}_1\cdot\boldsymbol{x}_2 + \gamma_1\gamma_2(\boldsymbol{b}\cdot\boldsymbol{x}_2)\boldsymbol{x}_1^2 + \gamma_1^2\boldsymbol{x}_2^2 - \gamma_1^2(\boldsymbol{b}\cdot\boldsymbol{x}_2)\boldsymbol{x}_2^2, \\
&\gamma_1\gamma_2(\boldsymbol{b}\cdot\boldsymbol{x}_1)\boldsymbol{x}_2^2 - \gamma_1\gamma_2\boldsymbol{b}^2\boldsymbol{x}_1^2\boldsymbol{x}_2^2 - \gamma_1^2(\boldsymbol{b}\cdot\boldsymbol{x}_2)\boldsymbol{x}_2^2 + \gamma_1^2\boldsymbol{b}^2\boldsymbol{x}_2^4.
\end{aligned} \tag{64}
$$

Note that

$$
\begin{aligned}
2\gamma_1\gamma_2(\boldsymbol{b}\cdot\boldsymbol{x}_1)\boldsymbol{x}_2^2 - \gamma_1\gamma_2\boldsymbol{b}^2\boldsymbol{x}_1^2\boldsymbol{x}_2^2 &= \gamma_1\gamma_2\boldsymbol{x}_2^2 - \gamma_1\gamma_2\boldsymbol{x}_2^2(1 - 2\boldsymbol{b}\cdot\boldsymbol{x}_1 + \boldsymbol{b}^2\boldsymbol{x}_1^2) \\
&= \gamma_1\gamma_2\boldsymbol{x}_2^2 - \gamma_1^2\gamma_2\boldsymbol{x}_2^2, \\
2\gamma_1\gamma_2(\boldsymbol{b}\cdot\boldsymbol{x}_2)\boldsymbol{x}_1^2 - \gamma_1\gamma_2\boldsymbol{b}^2\boldsymbol{x}_1^2\boldsymbol{x}_2^2 &= \gamma_1\gamma_2\boldsymbol{x}_1^2 - \gamma_1\gamma_2\boldsymbol{x}_1^2(1 - 2\boldsymbol{b}\cdot\boldsymbol{x}_2 + \boldsymbol{b}^2\boldsymbol{x}_2^2) \\
&= \gamma_1\gamma_2\boldsymbol{x}_1^2 - \gamma_1\gamma_2^2\boldsymbol{x}_1^2,
\end{aligned} \tag{65}
$$

so that $\beta^\mu\beta_\mu$ becomes

$$
\begin{aligned}
&\gamma_2^2\boldsymbol{x}_1^2 - \gamma_2^2(\boldsymbol{b}\cdot\boldsymbol{x}_1)\boldsymbol{x}_1^2 - \gamma_1\gamma_2\boldsymbol{x}_1\cdot\boldsymbol{x}_2 - \gamma_2^2(\boldsymbol{b}\cdot\boldsymbol{x}_1)\boldsymbol{x}_1^2 + \gamma_2^2\boldsymbol{b}^2\boldsymbol{x}_1^4 \\
&-\gamma_1\gamma_2\boldsymbol{x}_1\cdot\boldsymbol{x}_2 + \gamma_1^2\boldsymbol{x}_2^2 - \gamma_1^2(\boldsymbol{b}\cdot\boldsymbol{x}_2)\boldsymbol{x}_2^2 - \gamma_1^2(\boldsymbol{b}\cdot\boldsymbol{x}_2)\boldsymbol{x}_2^2 + \gamma_1^2\boldsymbol{b}^2\boldsymbol{x}_2^4 \\
&+\gamma_1\gamma_2\boldsymbol{x}_2^2 - \gamma_1^2\gamma_2\boldsymbol{x}_2^2 + \gamma_1\gamma_2\boldsymbol{x}_1^2 - \gamma_1\gamma_2^2\boldsymbol{x}_1^2 \\
=\ &(\gamma_1\gamma_2\boldsymbol{x}_1^2 + \gamma_1\gamma_2\boldsymbol{x}_2^2 - 2\gamma_1\gamma_2\boldsymbol{x}_1\boldsymbol{x}_2) + (\gamma_2^2\boldsymbol{x}_1^2 - \gamma_1\gamma_2^2\boldsymbol{x}_1^2 - 2\gamma_2^2(\boldsymbol{b}\cdot\boldsymbol{x}_1)\boldsymbol{x}_1^2 + \gamma_2^2\boldsymbol{b}^2\boldsymbol{x}_1^4) \\
&+(\gamma_1^2\boldsymbol{x}_2^2 - \gamma_1^2\gamma_2\boldsymbol{x}_2^2 - 2\gamma_1^2(\boldsymbol{b}\cdot\boldsymbol{x}_2)\boldsymbol{x}_2^2 + \gamma_1^2\boldsymbol{b}^2\boldsymbol{x}_2^4),
\end{aligned} \tag{66}
$$

and notice that

$$
\begin{aligned}
\gamma_1\gamma_2\boldsymbol{x}_1^2 + \gamma_1\gamma_2\boldsymbol{x}_2^2 - 2\gamma_1\gamma_2\boldsymbol{x}_1\boldsymbol{x}_2 &= \gamma_1\gamma_2(\boldsymbol{x}_1 - \boldsymbol{x}_2)^2, \\
\gamma_2^2\boldsymbol{x}_1^2 - \gamma_1\gamma_2^2\boldsymbol{x}_1^2 = \gamma_2^2\boldsymbol{x}_1^2(1 - \gamma_1) &= 2\gamma_2^2(\boldsymbol{b}\cdot\boldsymbol{x}_1)\boldsymbol{x}_1^2 - \gamma_2^2\boldsymbol{b}^2\boldsymbol{x}_1^4, \\
\gamma_1^2\boldsymbol{x}_2^2 - \gamma_1^2\gamma_2\boldsymbol{x}_2^2 = \gamma_1^2\boldsymbol{x}_2^2(1 - \gamma_2) &= 2\gamma_1^2(\boldsymbol{b}\cdot\boldsymbol{x}_2)\boldsymbol{x}_2^2 - \gamma_1^2\boldsymbol{b}^2\boldsymbol{x}_2^4,
\end{aligned} \tag{67}
$$

finally we proved that $\beta^\mu \beta_\mu = \gamma_1\gamma_2(\boldsymbol{x}_1 - \boldsymbol{x}_2)^2$ and $(\boldsymbol{x}_1' - \boldsymbol{x}_2')^2 = \frac{(\boldsymbol{x}_1 - \boldsymbol{x}_2)^2}{\gamma_1\gamma_2}$.

The constraint Equation 61 is satisfied only if $\Delta_1 = \Delta_2$. In other wards, two quasi-primary fields are correlated only if they have the same scaling dimesion:

$$\langle \phi_1(\boldsymbol{x}_1)\phi_2(\boldsymbol{x}_2)\rangle = \begin{cases} \frac{C_{12}}{|\boldsymbol{x}_1 - \boldsymbol{x}_2|^{2\Delta_1}} & \text{if} \quad \Delta_1 = \Delta_2 \\ 0 & \text{if} \quad \Delta_1 \neq \Delta_2 \end{cases} \tag{68}$$

And the 3-point function is similarly restricted. Invariance under translations,rotations, and dilatations requires

$$\langle \phi_1(\boldsymbol{x}_1)\phi_2(\boldsymbol{x}_2)\phi_3(\boldsymbol{x}_3)\rangle = \sum_{a,b,c} \frac{C_{abc}}{|\boldsymbol{x}_1 - \boldsymbol{x}_2|^a |\boldsymbol{x}_2 - \boldsymbol{x}_3|^b |\boldsymbol{x}_1 - \boldsymbol{x}_3|^c}, \tag{69}$$

and the summation over a, b, c is restricted such that $a + b + c = \Delta_1 + \Delta_2 + \Delta_3$, and special conformal transformations require $a = \Delta_1 + \Delta_2 - \Delta_3, b = \Delta_2 + \Delta_3 - \Delta_1, c = \Delta_3 + \Delta_1 - \Delta_2$. Thus, we get the final form of the 3-point correaltion function:

$$\langle \phi_1(\boldsymbol{x}_1)\phi_2(\boldsymbol{x}_2)\phi_3(\boldsymbol{x}_3)\rangle = \frac{C_{abc}}{|\boldsymbol{x}_1 - \boldsymbol{x}_2|^{\Delta_1+\Delta_2-\Delta_3} |\boldsymbol{x}_2 - \boldsymbol{x}_3|^{\Delta_2+\Delta_3-\Delta_1} |\boldsymbol{x}_1 - \boldsymbol{x}_3|^{\Delta_3+\Delta_1-\Delta_2}} \tag{70}$$

However, only 2-point and 3-point correlation functions have these neat forms, in general, n-point function will depend on the cross-ratio: $\frac{r_{ij}r_{kl}}{r_{ik}r_{jl}}$, for example, the 4-point correaltion function may take the following form:

$$\langle \phi_1(\boldsymbol{x}_1)\cdots\phi_4(\boldsymbol{x}_4)\rangle = f\left(\frac{r_{12}r_{34}}{r_{13}r_{24}}, \frac{r_{12}r_{34}}{r_{23}r_{14}}\right) \prod_{i<j}^{4} r_{ij}^{\frac{\Delta}{3}-\Delta_i-\Delta_j} \quad \text{where } \Delta = \sum_{i=1}^{4} \Delta_i. \tag{71}$$